

GML_04_04-capm.R

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# feb 2023
# r script for estimation of CAPM
# written by Frank Lehrbass (->lehrbass.de) for teaching purposes only
# students are solely responsible for whatever they do with this coding
# this R coding is WITHOUT ANY WARRANTY

#CAPM data from example in Verbeek book ch 2, pls read the stuff to get the
background
#also read Schroeder 2002, chapter on CAPM using DB and DAX!

#u may want to use this as template for "Determinanten des Aktienkurses von
XYZ"
#u may add Fama French factors
http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\_library.html#Research
#and add other X acc to your understanding of the business of XYZ

# Clean previous variables

rm(list=ls(all=TRUE))

# Load packages

library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method           from
##   as.zoo.data.frame zoo

library(lmtest)

## Warning: Paket 'lmtest' wurde unter R Version 4.3.3 erstellt

## Lade nötiges Paket: zoo

##
## Attache Paket: 'zoo'

## Die folgenden Objekte sind maskiert von 'package:base':
##
##   as.Date, as.Date.numeric

library(car)

## Warning: Paket 'car' wurde unter R Version 4.3.3 erstellt
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## Lade nötiges Paket: carData

library(sandwich)

## Warning: Paket 'sandwich' wurde unter R Version 4.3.3 erstellt

library(rugarch)

## Lade nötiges Paket: parallel

##
## Attache Paket: 'rugarch'

## Das folgende Objekt ist maskiert 'package:stats':
##
##      sigma

data_table <- read.table("GML_04_04 capm.csv", header=TRUE, sep=";", dec=",")

summary(data_table)

##      month      constrrf      durblrf      foodrf
## Min.      :196001  Min.      :-29.8100  Min.      :-25.9000  Min.      :-18.800
## 1st Qu.:197209  1st Qu.: -3.1800  1st Qu.: -2.8500  1st Qu.: -1.603
## Median :198506  Median :  0.4200  Median :  0.4250  Median :  0.715
## Mean    :198498  Mean    :  0.4379  Mean    :  0.3275  Mean    :  0.653
## 3rd Qu.:199802  3rd Qu.:  3.7475  3rd Qu.:  4.0000  3rd Qu.:  3.223
## Max.    :201010  Max.    : 25.5200  Max.    : 29.4500  Max.    : 19.520
##      hml      jan      rf      rmrf
## Min.      :-12.7800  Min.      :0.00000  Min.      :0.000  Min.      :-23.140
## 1st Qu.: -1.1500  1st Qu.:0.00000  1st Qu.:0.280  1st Qu.: -2.205
## Median :  0.4350  Median :0.00000  Median :0.410  Median :  0.840
## Mean    :  0.4022  Mean    :0.08361  Mean    :0.427  Mean    :  0.437
## 3rd Qu.:  1.7975  3rd Qu.:0.00000  3rd Qu.:0.530  3rd Qu.:  3.462
## Max.    : 13.8400  Max.    :1.00000  Max.    :1.350  Max.    : 16.050
##      smb
## Min.      :-16.670
## 1st Qu.: -1.460
## Median :  0.070
## Mean    :  0.230
## 3rd Qu.:  2.038
## Max.    : 22.190

#your data should be daily and at least 2 yrs, more is better but watch out
for structural breaks
#note that Verbeek uses excess returns with rf = return on 1 mth T Bills
#WPs (auditors, corp valuation) Love a Longer rate eg Treasuries 20yrs

# Diagnostics of one specific regression

#adjust this section!!!=====
Y = data_table$constrrf #xcess return for a sub-index of industrials, cap
weighted like dax
```

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X = data_table$rmrf #xcess for broad CRSP index
reg = lm(Y~X) #play here if u want
summary(reg)

##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.879  -1.641  -0.115   1.520  11.725
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.07259    0.11542  -0.629    0.53
## X           1.16817    0.02549  45.837 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.837 on 608 degrees of freedom
## Multiple R-squared:  0.7756, Adjusted R-squared:  0.7752
## F-statistic: 2101 on 1 and 608 DF, p-value: < 2.2e-16

# Estimate CAPM with GARCH for Construction

data_ext<-matrix(data= NA , nrow=length(residuals(reg)), ncol=2)
data_ext<-cbind(data_table$constrrf,data_table$rmrf)
data_ext=rbind(data_ext,data_ext)

spec2 <- ugarchspec(variance.model = list(model = "sGARCH", garchOrder = c(1,
1),
                                variance.targeting = TRUE),
                    mean.model = list(armaOrder = c(0, 0), include.mean =
TRUE,
                                external.regressors =
matrix(data_ext[,2])),
                    distribution.model = "norm")
garch2 <-
ugarchfit(spec=spec2,data=matrix(data_ext[,1]),solver.control=list(trace=0))

coef(garch2)

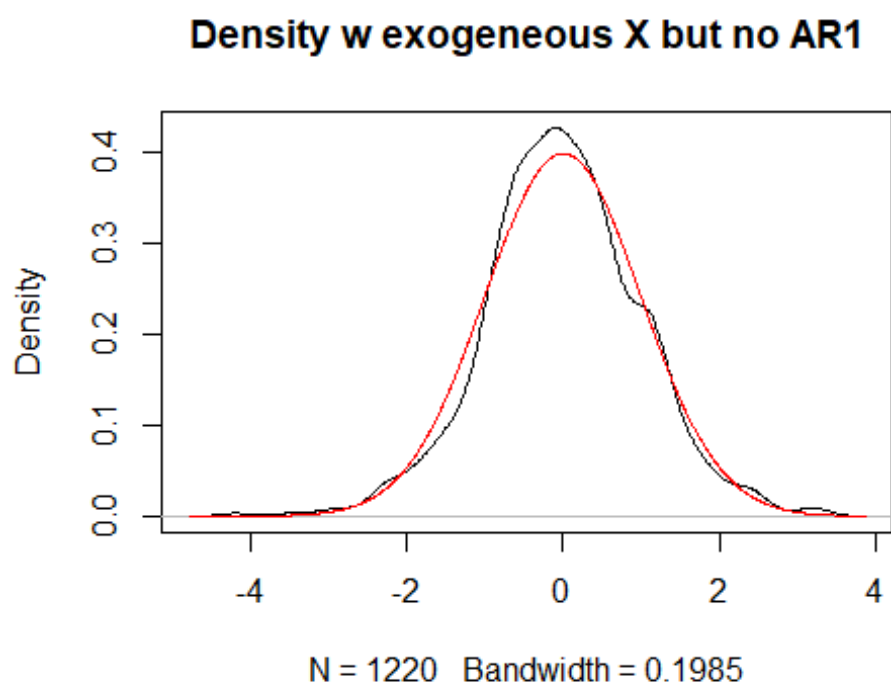
##           mu      mxreg1      alpha1      beta1      omega
## -0.13517802  1.19623699  0.07863426  0.89519503  0.21049941

#compare and note that beta is now slightly higher than in ordinary OLS
coefficients(reg)

## (Intercept)           X
## -0.07258514  1.16816828

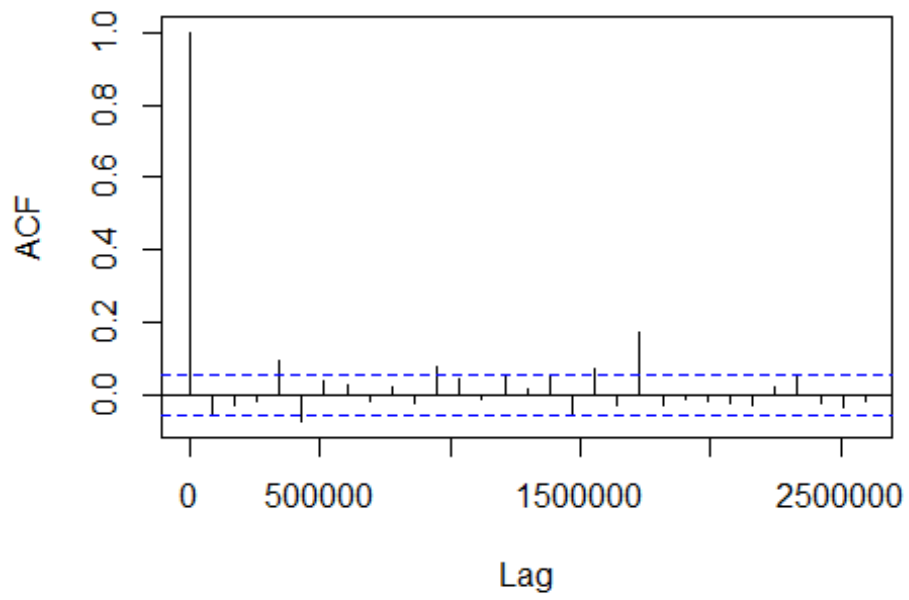
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```
std_res2 = residuals(garch2)/sigma(garch2)
kernel_den = density(std_res2)
x11()
plot(kernel_den, main="Density w exogeneous X but no AR1")
x <- seq(min(kernel_den$x),max(kernel_den$x),length = 1000)
normal_den = dnorm(x,mean = mean(std_res2),sd = sd(std_res2))
lines(x,normal_den,type="l",col="red")
```



```
x11() #note that ARCH effects captured now in model
acf(std_res2^2)
```

Series std_res2^2



#end

#Ende